```python
def related_samples_ttest(diff):

    values = list(diff)
    n = len(values)

    mean_diff = sum(values) / n
    std_diff = math.sqrt(sum((x - mean_diff) ** 2 for x in values) / (n - 1))

    df = n - 1# degrees of freedom
    se = std_diff / math.sqrt(n) # standard error of the mean difference

    t_stat, p_value = ttest_1samp(values, popmean=0)

    t_critical = t.ppf(0.975, df)
    ci_lower = mean_diff - t_critical * se
    ci_upper = mean_diff + t_critical * se

    print("     Summary Statistics")
    print(f"  N                    : {n}")
    print(f"  Mean Difference      : {mean_diff:.4f}")
    print(f"  Std. Dev of Difference   : {std_diff:.4f}")
    print(f"  Std. Error Mean      : {se:.4f}")
    print(f"  t-statistic          : {t_stat:.4f}")
    print(f"  Degrees of Freedom (df)  : {df}")
    print(f"  Confidence Interval  : ({ci_lower:.4f}, {ci_upper:.4f})")
    print(f"  p-value       : {p_value:.4f}")
    print("=" * 55)
    print()
```

Printing column labels:
Column labels: ['Sex', 'September Weight', 'April Weight', 'September BMI', 'April BMI']

```python
# Printing column labels using pandas
print("Column labels:", df.columns.tolist())
print()
```

New column and checking if it's correct:

```python
df['Weight Difference'] = df['September Weight'] - df['April Weight']
```

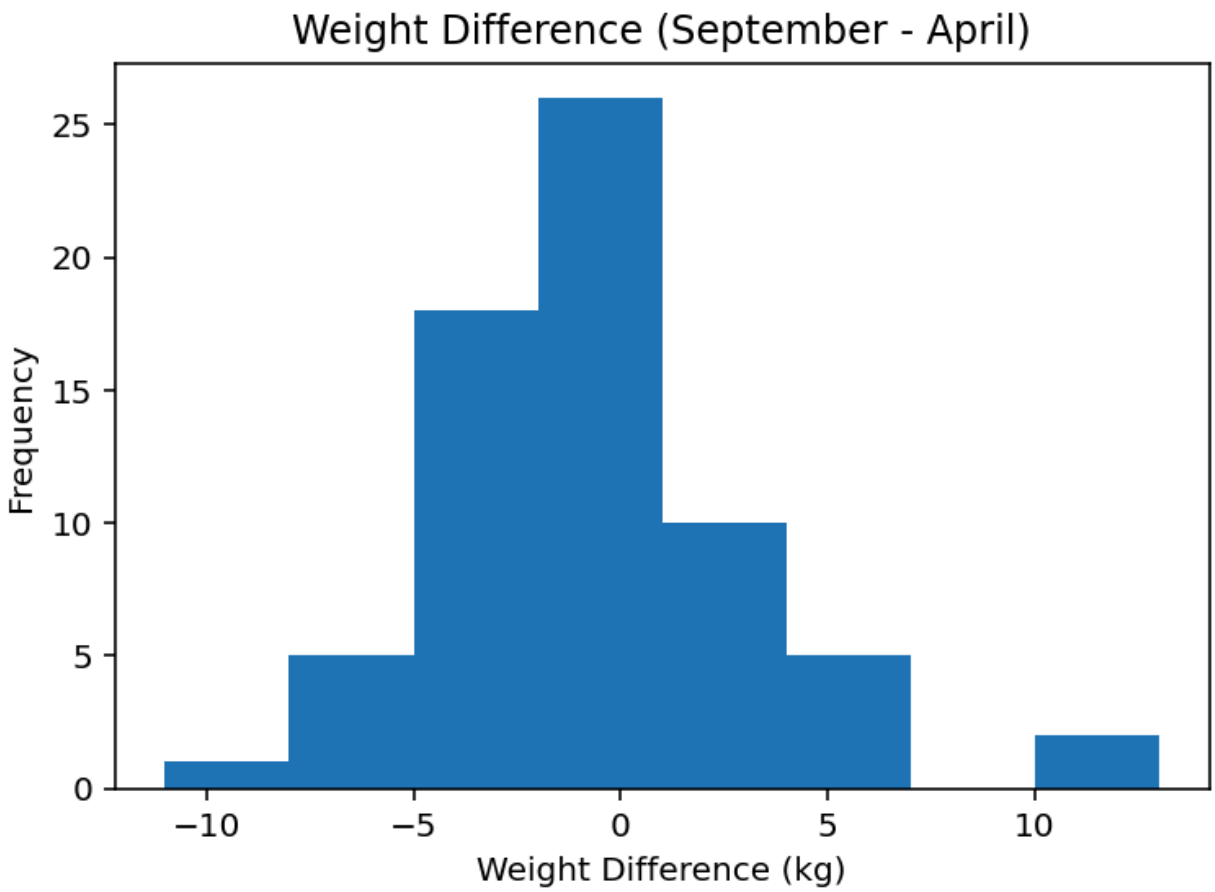
This printed the columns

```python
# check the column looks correct
print("First 5 rows - checking Weight Difference:")
print(df[['September Weight', 'April Weight', 'Weight Difference']].head())
print()
```

And they are correct

```
First 5 rows - checking Weight Difference:
   September Weight  April Weight  Weight Difference
0                72            59                 13
1                97            86                 11
2                74            69                  5
3                93            88                  5
4                68            64                  4
```

Weight difference histogram:



Weight difference statistics

The mean weight difference (September − April) was −1.18 kg (N = 67, SD = 3.89). A related samples t-test gave t(66) = −2.48, p = .016. Because p < .05, we reject the null hypothesis. There is a significant difference in weight between September and April. The negative mean mean that students weighed less in September than in April, meaning they gained weight over the course of the school year.

For male participants only (N = 32), the mean weight difference was −1.16 kg (SD = 4.91). The related samples t-test was t(31) = −1.33, p = .193. Because p > .05, we fail to reject the null hypothesis. There is no statistically significant difference in weight between September and April for male students alone. The result suggests that when looked at separately, males did not show a significant weight change during the school year

```
--- T-Test: Weight Difference (Males only) ---
    Summary Statistics
  N                        : 32
  Mean Difference          : -1.1562
  Std. Dev of Difference   : 4.9128
  Std. Error Mean          : 0.8685
  t-statistic              : -1.3314
  Degrees of Freedom (df)  : 31
  Confidence Interval      : (-2.9275, 0.6150)
  p-value        : 0.1928
```

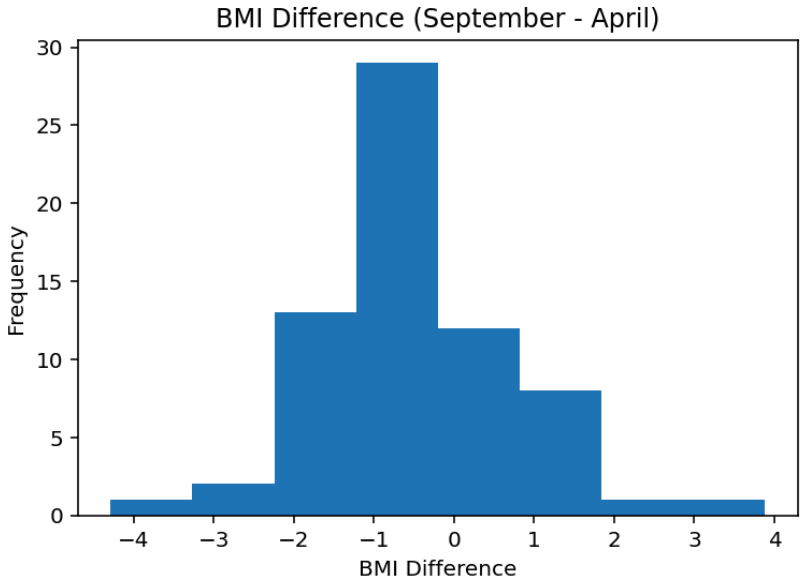
BMI Difference

For male participants only (N = 32), the mean BMI difference (September − April) was −0.44 (SD = 1.51). The related samples t-test gave t(31) = −1.66, p = .107. Because p > .05, we fail to reject the null hypothesis.

```
--- Freshman 15 Dataset: BMI ---

BMI Difference (All):
N = 67
Mean = -0.45417910447761184
Std Dev = 1.2595371314960775

--- T-Test: BMI Difference (Males only) ---
    Summary Statistics
  N                        : 32
  Mean Difference          : -0.4416
  Std. Dev of Difference   : 1.5064
  Std. Error Mean          : 0.2663
  t-statistic              : -1.6582
  Degrees of Freedom (df)  : 31
  Confidence Interval      : (-0.9847, 0.1015)
  p-value        : 0.1074
```



3. Parent Heights

H_0: The mean difference between child height and mid-parent height = 0.

H_1: The mean difference between child height and mid-parent height ≠ 0.

```
Child vs Mid-Parent Height Difference:
N = 40
Mean = -0.6999999999999995
Std Dev = 3.0342489460600732

--- T-Testt ---
     Summary Statistics
  N                          : 40
  Mean Difference            : -0.7000
  Std. Dev of Difference     : 3.0342
  Std. Error Mean            : 0.4798
  t-statistic                : -1.4591
  Degrees of Freedom (df)    : 39
  Confidence Interval        : (-1.6704, 0.2704)
  p-value          : 0.1526
```

The mean difference (child height − mid-parent height) was −0.70 inches (N = 40, SD = 3.03). The histogram shows the differences. The related samples t-test gave t(39) = −1.46, p = .153. Because p > .05, we fail to reject the null hypothesis. The hypothesis was not supported children's heights are not significantly different from the average of their parents' heights.

